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Bachelor of Liberal Arts & Sciences

SCIMATH202 Theory of Statistics

Project Report

Random Number Generating Algorithms For Sampling Various Distributions

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1 Introduction

The field of random number generation algorithms describes procedures for generating independent observations from a particular standard distribution, such as the Normal or Poisson distribution. These random samples can be used as an important tool for various applications in fields such as, statistics, physics, computer science and engineering.

Most modern computer software packages are already equipped with functions that can generate random samples from a whole variety of distributions. However, it is still useful and interesting to know how these functions exactly generate these samples, since the choice of sampling algorithm or function depends on the specific requirements of the desired distribution and the limitations of each algorithm.

Note that most software packages are also capable of generating pseudo-random real numbers. These are real numbers that seem to be random but are actually generated using deterministic procedures, as this is the way for computers to operate. However, from a practical point of view, these numbers can be treated to be truly random. Taking advantage of this property, it is possible to create an Uniform distribution with a range between 0 and 1. The random number generating algorithms describe a set of steps to transform this created random Uniform distribution into the distribution that is desired [1].

Some standard distributions, such as the Bernoulli and Binomial, can be derived by Additive sampling methods: iteratively adding or subtracting a simple transformation of the basic distribution that is easier to sample from, i.e., the uniform distribution in this case. However, this approach does not work for every distribution, and for these cases other methods are needed. Common random number-generating algorithms that can be used for these distributions include the Inverse Transform Method, the Box-Mueller Transform, and the Accept and Reject Method. The explanation and limitations behind each of these methods will be discussed in this report along with a number of examples to show their performance which are simulated in *Mathematica*.

2 Additive Sampling

As described in the introduction, section 1, the basic idea behind additive sampling is to add or subtract a simple transformation of a base distribution over and over until it gets close enough to the desired distribution. In the following sections, a couple of examples will be discussed to show the performance of this set of algorithms.

2.1 Bernoulli

Suppose that it is needed to generate a sample of the Bernoulli distribution with the probability for success and failure being p and $q = 1 - p$ respectively. As mentioned before, it is relatively straightforward to generate a sample of the uniform distribution, by just generating random real numbers between 0 and 1. Define the random variable X and $U \sim Uniform(0, 1)$. Now set $X = 0$ for $U < p$ and $X = 1$ for $U > p$ Doing this will transform X into a Bernoulli distribution with success parameter p. An example of a sampled Bernoulli distribution with $p = 0.65$ is shown in figure 1 below, by creating 50 000 observation using the steps described above.

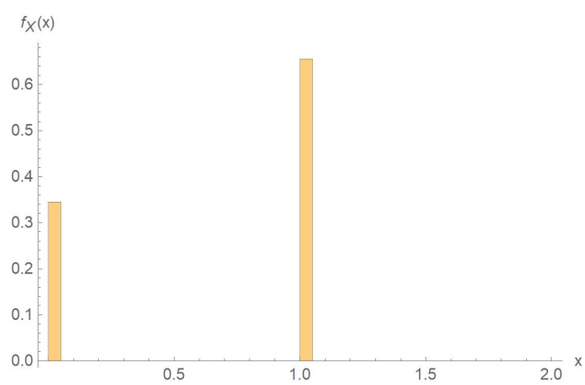

Figure 1: Bernoulli Distribution with $p = 0.65$

2.2 Binomial

Now it is possible to generate a Bernoulli sample, the next step is to get a sample of the Binomial distribution. A relation between the Bernoulli and Binomial distribution can be formulated through their moment-generating functions (MGF). The MGF of the random variable $X \sim Bernoulli(p)$ is given as

$$m_X(t) = \varepsilon[e^{tX}] = q + pe^t \tag{1}$$

and that of the random variable $Y \sim Binomial(n, p)$, where n is the number of Bernoulli trials, is

$$m_Y(t) = \varepsilon[e^{tY}] = \left(q + pe^t\right)^n . \tag{2}$$

Now generate the n identically and independent random variables $X_1 = X_2 =, \ldots, = X_n \sim Bernoulli(p)$ and set $Z = X_1 + X_2 \cdots + X_n$. The MGF of Z follows as

$$\begin{aligned} m_Z(t) &= \varepsilon[e^{t(X_1+X_2+\cdots+X_n)}] \\ &= \varepsilon[e^{tX_1}]\varepsilon[e^{tX_2}]\ldots\varepsilon[e^{tX_n}] \\ &= \left(q+pe^t\right)\left(q+pe^t\right)\cdot\cdots\cdot\left(q+pe^t\right) \\ &= \left(q+pe^t\right)^n \end{aligned} \tag{3}$$

As can be seen, Eq. (2) and (3) are identical. From this result, it can be concluded that a Binomial distribution with parameters n and p can be generated by summing up the random variables $X_1, X_2 \ldots, X_n$, which can be generated using the theorem explained in section 2.1 [2]. An example of this implementation, with $p = 0.65$ and $n = 50$, is given in figure 2 below.

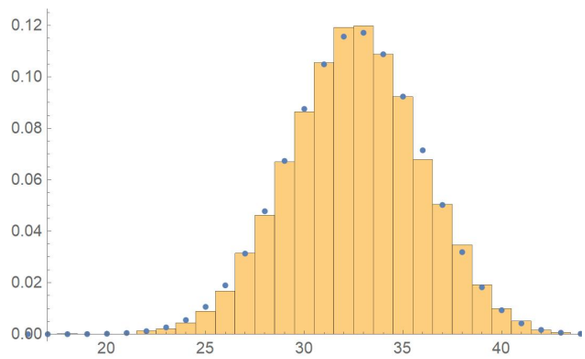

Figure 2: Histogram of sampled Binomial distribution against the analytical values ($p = 0.65$ and $n = 50$)

2.3 Geometric

In addition to the Binomial distribution, a sample of the Geometric distribution can now be generated using the sampling method for a Bernoulli trial. The Geometric distribution defines the probability of the number of failures occurring prior to the first success in a repeated game of Bernoulli trials.

To generate a Geometric distribution, a series of n $Bernoulli(p)$ variables, $Y_1 \ldots Y_n$ will be generated using the method described in section 2.1, and their outcomes are stored in an array. After this the random variable $Y = I - 1$ is defined, where I is the index of the first successful Bernoulli sample, such that $Y_I = 1$ [2]. This method is repeated a relatively large number of times (e.g. 50 000) and the resulting distribution for one example is shown in figure 3 below.

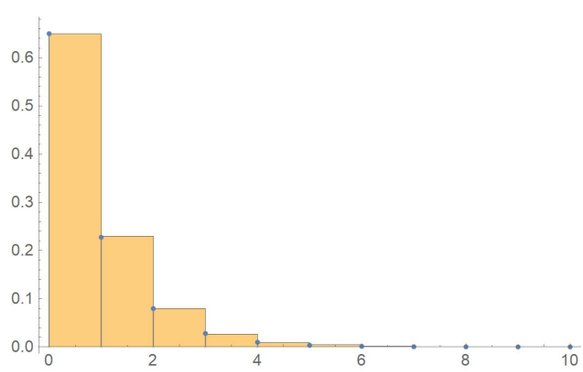

Figure 3: Histogram of the sampled Geometric distribution against the analytical values ($p = 0.65, n = 50$)

2.4 Negative Binomial

The Negative Binomial and the Geometric distributions are related in a similar way as the Binomial and Bernoulli distribution. The MGF of a random variable $X \sim Neg.Binomial(p, r)$, where X denotes the number of failures prior to the $r - th$ success, is denoted as [1, 2]

$$m_X(t) = \left(\frac{p}{1-qe^t}\right)^r. \tag{4}$$

Now generate the r identically and independent random variables $X_1 = X_2 =, \ldots, = X_r \sim Geometric(p)$

and set $Y = X_1 + X_2 \cdots + X_r$. The MGF of Y follows as

$$\begin{aligned} m_Y(t) &= \varepsilon[e^{t(X_1+X_2+\cdots+X_r)}] \\ &= \varepsilon[e^{tX_1}]\varepsilon[e^{tX_2}]\ldots\varepsilon[e^{tX_r}] \\ &= \left(\frac{p}{1-qe^t}\right)_1 \left(\frac{p}{1-qe^t}\right)_2 \cdots \left(\frac{p}{1-qe^t}\right)_r \cdot \\ &\left(\frac{p}{1-qe^t}\right)^r \end{aligned} \tag{5}$$

As can be seen , Eq. (2) and (4) are identical. From this result, it can be concluded that a Negative Binomial distribution with parameters r and p can be generated by summing up the random variables $X_1, X_2 \ldots, X_r$, which can be generated using the theorem explained in section 2.3. An example of this implementation, with $p = 0.65$ and $r = 50$, is given in figure 4 below.

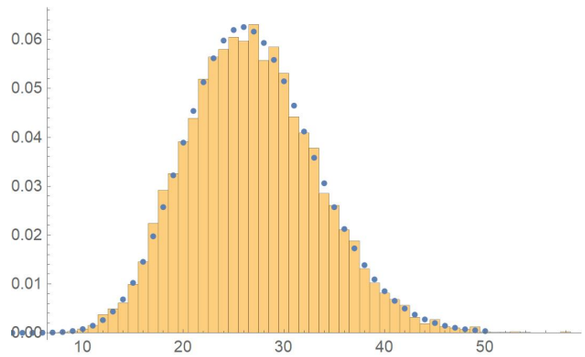

Figure 4: Histogram of the sampled Negative Binomial distribution against the analytical values ($p = 0.65$, $r = 50$)

2.5 Hypergeometric

The Hypergeometric distribution describes the probability of the number of successes occurring in a sample of a fixed size (n) drawn without replacement from a population (N) that consists out of a number of success (K) and failure ($N - K$) elements. This distribution can be used in examples where there is a fluent probability for success at each trial. The corresponding PDF of the random variable X with this distribution is defined as

$$f_X(x) = \frac{\begin{pmatrix} K \\ x \end{pmatrix}\begin{pmatrix} N-K \\ n-x \end{pmatrix}}{\begin{pmatrix} N \\ n \end{pmatrix}}, \tag{6}$$

where x denotes the number of success items drawn in the sample. The terms in the numerator denote the number of combinations possible to draw x successes and $n - x$ failures from the population N. The term in the denominator represent the total number of combinations to take a sample of size n from the population of N. Dividing these two terms over each other will thus give the probability of choosing x success elements of sample n out of the population N [3].

The Binomial and Hypergeometric distributions are closely related, with the main difference being that in the first distribution the probability for success remains constant for each trial, while this is not the case for the latter. However, if the size of the population N is much larger then the sample size n, this will mean that the outcome of one trial has almost no effect on the probability that the next selected element is a success or failure. Using this condition, a sample of the Hypergeometric can be approximated by using the same steps as for generating a Binomial sample as described in section 2.2. The random variable $Y = X_1 + \cdots + X_n$, where $X_i \sim Bernoulli(p)$ and $p = \frac{K}{N}$, will follow a Binomial distribution with parameters p and n, and this will be a sufficient approximation for the Hypergeometric distribution with parameters N, K and n, if $N >> n$. The value for p is kept as a constant, and since N (and K) is a relatively large number, this assumption will be acceptable [3,4]. Figure 5 below shows the working of this method for a specific example.

3 Inverse Transform Method

The Inverse Transform Method (ITM) is another algorithm that can be used for generating random samples from a particular distribution, given that the cumulative density function (CDF) of this distribution can be defined. The derivation of this method starts by considering the random variable X that follows the probability distribution of $f_X(x)$ and its CDF is defined as

$$F_X(x) = P(X \leq x) = y, \tag{7}$$

where y is some percentile and corresponds to the value to value of the random variable Y.
Now the goal is to find the CDF of Y, which is defined as the CDF of the random variable X. Therefore,

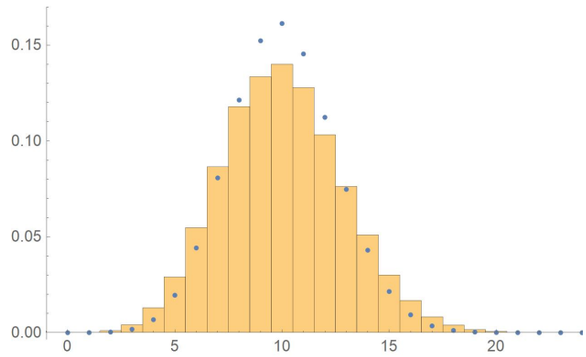

Figure 5: Histogram of the sampled Hypergeometric distribution against the analytical values ($N = 200, K = 40, n = 50$)

$$F_Y(y) = P(Y \leq y) = P(F_X(x) \leq y). \tag{8}$$

Now taking the inverse CDF on both sides of the probability condition yields

$$F_Y(y) = P(F_X^{-1}(F_X(x)) \leq F_X^{-1}(y)). \tag{9}$$

Note that this only holds when the random variable X has a well defined inverse CDF. This is the case when the CDF is strictly increasing. It is rather trivial to see that the inverse CDF of the CDF is simply the value of the random variable X. Therefore, the above equation simplifies to

$$F_Y(y) = P(x \leq F_X^{-1}(y)). \tag{10}$$

Note that Eq. (10) can be recognized as the CDF of the random variable x in terms of $F_X^{-1}(y)$, as is shown in Eq. (7). This implies that

$$F_Y(y) = F_X^{-1}(F_X(y)) = y \tag{11}$$

However, an interesting result is the corresponding PDF of the random variable Y:

$$f_Y(y) = \frac{d}{dy}F_Y(y) = \frac{d}{dy}y = 1. \tag{12}$$

This means that the random variable Y follows an Uniform distribution. Rewriting Eq. (7) in terms of x gives:

$$x = F_X^{-1}(y). \tag{13}$$

In conclusion, observations from the desired random variable X can be sampled by taking the inverse CDF of this variable and inserting observations from the uniform distribution into this equation [5]. This method is relatively efficient as it just transforms observations from the uniform distribution into any other desired distribution. However, its shortcoming is that the CDF of some PDFs can be hard to define. In the upcoming sections a couple of examples will be shown how the ITM can be used solely, aswell in combination with the Additve sampling method [6].

3.1 Triangular

The Triangular distribution is a distribution with a PDF, as the name already suggests, in the form of a triangle and is defined by

$$f(x; a, b, c) = \begin{cases} \frac{2}{(c-a)(b-a)}\,(x-a)\,, & a \leq x < b \\ -\frac{2}{(c-a)(c-b)}\,(x-c)\,, & b \leq x \leq c \\ 0 \qquad\qquad\qquad , & otherwise \end{cases}, \tag{14}$$

where $a < b < c$ [7]. To use the ITM, the CDF has to be defined by integrating the non-zero pieces of Eq. (14). The CDF on $a \leq x < b$ is

$$\begin{aligned} \int_{u=a}^{x} \frac{2}{(c-a)(b-a)}(u-a)du &= \left[\frac{(u-a)^2}{(c-a)\,(b-a)}\right]_a^x = \frac{(x-a)^2}{(c-a)\,(b-a)} - \frac{(a-a)^2}{(c-a)\,(b-a)} \\ &= \frac{(x-a)^2}{(c-a)\,(b-a)} \qquad , \end{aligned} \tag{15}$$

and on $b \leq x \leq c$

$$\begin{aligned}
&\int_{u=a}^{b} \frac{2}{(c-a)(b-a)}(u-a)du + \int_{u=b}^{x} -\frac{2}{(c-a)\,(c-b)}\,(u-c)\,du \\
&= \frac{(b-a)^2}{(c-a)\,(b-a)} - \left[\frac{(u-c)^2}{(c-a)(c-b)}\right]_b^x \\
&\frac{(b-a)^2}{(b-a)(c-a)} - \left(\frac{(x-c)^2}{(c-a)(c-b)} - \frac{(b-c)^2}{(c-a)(c-b)}\right) \\
&\frac{(b-a)^2(c-b) + (b-a)(b-c)^2}{(b-a)(c-a)(c-b)} - \frac{(x-c)^2}{(c-a)(c-b)} \\
&= 1 - \frac{(x-c)^2}{(c-a)(c-b)} \\
&= 1 - \frac{(c-x)^2}{(c-a)(c-b)} \qquad .
\end{aligned} \tag{16}$$

To avoid confusion the dummy variable u is introduced, since x already defines the upper bound of the integration range. The CDF is thus

$$F_X(x) = \begin{cases} 0, & x < a \\ \frac{(x-a)^2}{(c-a)(b-a)}, & a \leq x < b \\ 1 - \frac{(c-x)^2}{(c-a)(c-b)}, & b \leq x < c \\ 1, & x \geq c \end{cases}, \tag{17}$$

and $F_X^{-1}(y)$ can be found by setting Eq. (17) equal to y and then solving the equation for y. For x on $a \leq x < b$ this becomes

$$\begin{aligned}
\frac{(x-a)^2}{(c-a)\,(b-a)} &= y \\
(x-a)^2 &= y\,(c-a)\,(b-a) \\
x - a &= \sqrt{y\,(c-a)\,(b-a)} \\
x &= \sqrt{y\,(c-a)\,(b-a)} + a
\end{aligned} \tag{18}$$

and on $b \leq x < c$

$$\begin{aligned}
1 - \frac{(c-x)^2}{(c-a)(c-b)} &= y \\
\frac{(c-x)^2}{(c-a)(c-b)} &= 1 - y \qquad . \\
(c-x)^2 &= (c-a)\,(c-b)\,(1-y) \\
x &= c - \sqrt{(c-a)\,(c-b)\,(1-y)}
\end{aligned} \tag{19}$$

Inserting the boundary values of F_X into the equation of y will yield the boundary values of $F_X^{-1}(y)$. Therefore, $a \leq x < b$ becomes

$$\begin{aligned}
y &= \frac{(x-a)^2}{(c-a)\,(b-a)}, \quad x = a \\
&= 0 \\
y &= \frac{(x-a)^2}{(c-a)\,(b-a)}, \quad x = b \\
&= \frac{(b-a)^2}{(c-a)\,(b-a)} = \frac{b-a}{c-a}\ ,
\end{aligned} \tag{20}$$

and $b \leq x < c$

$$\begin{aligned}
y &= 1 - \frac{(c-x)^2}{(c-a)(c-b)}, \quad x = b \\
&= 1 - \frac{(c-b)^2(c-a)}{(c-b)} = 1 - \frac{(c-b)}{(c-a)} \\
&= \frac{(c-a-c+b)}{(c-a)} = \frac{b-a}{c-a} \\
y &= 1 - \frac{(c-x)^2}{(c-a)(c-b)}, \quad x = c \\
&= 1 \qquad .
\end{aligned} \tag{21}$$

Thus, the inverse CDF is defined as

$$F_X(y)^{-1} = \begin{cases} \sqrt{y\,(c-a)\,(b-a)} + a, & 0 \leq y < \frac{b-a}{c-a} \\ c - \sqrt{(c-a)\,(c-b)\,(1-y)}, & \frac{b-a}{c-a} \leq y \leq 1 \end{cases}. \tag{22}$$

Using this result, it is possible to transform observations of the uniform distribution into that of the triangular distribution by inserting these observations into Eq. (22) [8]. Figure 6 shows the obtained sample from 50 000 observations.

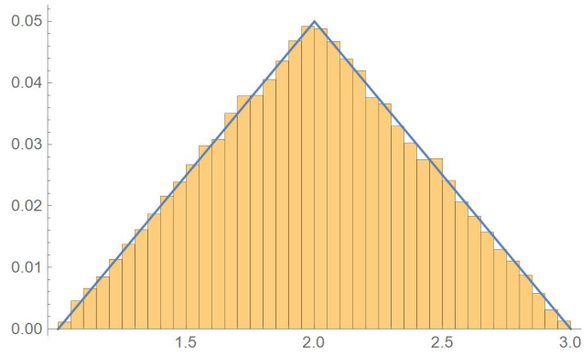

Figure 6: Histogram of the sampled Triangular distribution against the analytical PDF ($a = 1$, $b = 2$, $c = 3$)

3.2 Poisson and Exponential

The Poisson distribution models the probability of the number of times an event occurs in a fixed time interval when the events occur statistically independent of each other. This distribution only has one single parameter, λ, which denotes the average number of events happening in a specified time interval. The PDF of a random variable $X \sim Poisson(\lambda)$ is defined as

$$f_X(x) = \frac{e^{-\lambda}\lambda^x}{x!}, \tag{23}$$

where x is the number of Poisson events occurring in the time interval. Note that this distribution is discrete and is thus only defined for integer values of X [9].

Looking at this PDF, it seems to be rather difficult to use the ITM to generate a sample of this distribution as this function includes some terms that are not as straightforward to integrate. Therefore, it might be easier to first look at how to generate a sample of a distribution closely related to that of a Poisson process, namely the Exponential distribution.

The Exponential distribution is closely related to these Poisson processes, as it can be used to compute the probability that the first occurrence of a Poisson process will occur within a certain time interval. The PDF of this distribution is

$$f_X(x;\lambda) = \lambda e^{-\lambda x} \quad . \tag{24}$$

The CDF of this function is derived as

$$\begin{aligned} F_X(x;\lambda) &= \int_0^x \lambda e^{-\lambda u} du = \lambda \int_0^x e^{-\lambda u} du \\ &= \lambda \left[-\frac{1}{\lambda} e^{-\lambda u} \right]_0^x = -\left[e^{-\lambda u} \right]_0^x \\ &= -\left(e^{-\lambda x} - \left(e^{-\lambda 0} \right) \right) = 1 - e^{-\lambda x} \end{aligned} \tag{25}$$

Now the desired inverse CDF can be computed by setting the CDF equal to y and then solve the equation for y:

$$\begin{aligned} 1 - e^{-\lambda x} &= y \\ e^{-\lambda x} &= 1 - y \\ x &= -\frac{\ln(1-y)}{\lambda} \\ F_X(y)^{-1} &= -\frac{\ln(1-y)}{\lambda}. \end{aligned} \tag{26}$$

It is now possible to transform a random variable with the distribution of $U(0,1)$ into a sample of the Exponential distribution by inserting the observations from U into Eq. (26). Figure 7 below shows an example of a generated sample using this method.

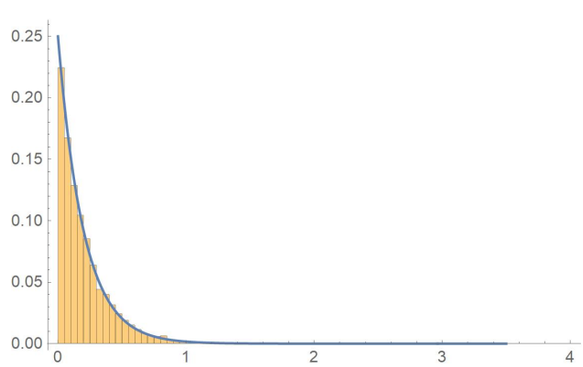

Figure 7: Histogram of the sampled Exponential distribution against the analytical PDF ($\lambda = 3$)

Given that it is now possible to generate a sample of the Exponential distribution, a sample of the Poisson distribution can be created using the following steps:

1. Generate a sample of the random variable $X \sim EXP(\lambda)$, consisting out of a relatively large number of observations and store the results in an array, using the explanation shown above Note, that X defines the time interval of the first Poisson event happening.
2. Denote T as the accumulation of the values of X, i.e. $T[n] = X[n] + X[n-1] + X[n-2] + \cdots + X[1]$. Now, $T[n]$ denotes the timestamp at which the n-th Poisson event is taking place.
3. As mentioned before, the Poisson distribution is discrete and is not defined for non-integer values. Therefore, the values of T are rounded down to the nearest integer value.
4. Finally, group the number of Poisson events taking place at each integer value and take the sum of these groups.

Using this algorithm, the random variable T will follow a Poisson distribution with rate parameter λ. An example, and also the verification, of this method is shown in figure 8 below.

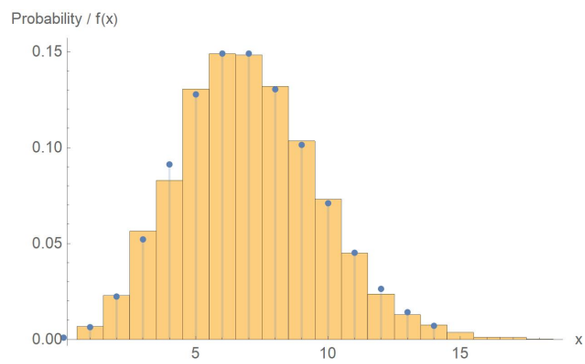

Figure 8: Histogram of the sampled Poisson distribution against the analytical PDF ($\lambda = 7$)

3.3 Gamma

Just as the Exponential distribution, the Gamma distribution is also closely related to the Poisson distribution, as it can be used to compute the probability that the r-th Poisson event occurs in a specified time interval. The PDF of this distribution is usually defined as

$$f(x;\alpha,\beta) = \frac{\beta^\alpha x^{\alpha-1} e^{-\beta x}}{\Gamma(\alpha)} \quad x,\alpha,\beta > 0 \tag{27}$$

or

$$f(x;k,\theta) = \frac{1}{\Gamma(k)\theta^k} x^{k-1} e^{-\frac{x}{\theta}}, \tag{28}$$

where $\alpha = k$ and β are the shape and rate parameter respectively, and $\theta = \frac{1}{\beta}$ is the scale parameter [10]. Furthermore, $\Gamma(\alpha)$ is the Gamma function that is formulated by

$$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx. \tag{29}$$

Computing the CDF of the Gamma distribution is rather challenging compared to the previous examples since integrating this PDF is not as straightforward.

However, note that the Gamma and Exponential distributions are closely related since the Exponential distribution is a special case of the Gamma distribution where $\alpha = 1$ and $\beta = \lambda$ [11].

Furthermore, a random sample of size n taken from $X \sim EXP(\lambda)$ has a sample total $\sum_{i=1}^n X_i = Y$, where X_i are identically and independent samples, that corresponds to the Gamma distribution. The proof of this theorem can be found in the moment-generating functions (MGF) of both distributions. Recall that the MGF of the random variable X will be

$$m_X(t) = \varepsilon[e^{tX}] = \frac{\lambda}{\lambda - t} \tag{30}$$

, which means that the MGF of Y is

$$\begin{aligned} m_Y(t) = \varepsilon[e^{t(X_1+X_2+\cdots+X_n)}] &= \varepsilon[e^{tX_1}]\varepsilon[e^{tX_1}]\ldots\varepsilon[e^{tX_n}] \\ &= m_{X_1}(t)m_{X_1}(t)\ldots m_{X_n}(t) \\ &= m_X^n(t) = \left(\frac{\lambda}{\lambda - t}\right)^n \end{aligned} \tag{31}$$

Furthermore, the MGF of the Gamma distribution is defined as [9]

$$m_Y(t) = \varepsilon[e^{tY}] = \left(1 - \frac{t}{\beta}\right)^{-\alpha} = \left(\frac{\beta - t}{\beta}\right)^{-\alpha} = \left(\frac{\beta}{\beta - t}\right)^{\alpha}. \tag{32}$$

The two equations for the MGF of Y are equal when the parameters are set to be $\alpha = n$ and $\beta = \lambda$. Thus $Y \sim Gamma(n, \lambda)$.

Using this knowledge it is possible to generate a sample from the Gamma distributions by following the next steps:

1. Generate an Exponential distribution as described in sec. 3.2 using the ITM.
2. Then, take n samples with a specified size of this distribution.
3. Take the total sum of each sample. This will give a list of n sample totals.
4. Put together, this list of n sample totals will thus yield a Gamma distribution with parameters n and λ.

Figure 9 below gives an example of a sampled Gamma distribution using this method.

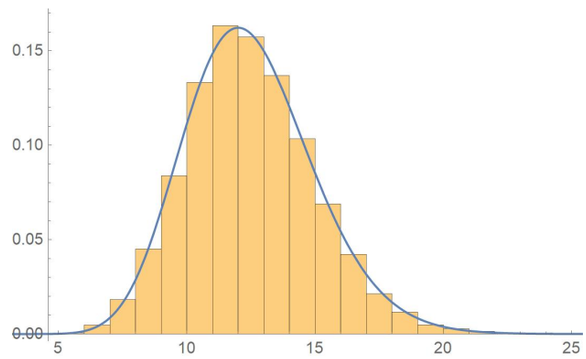

Figure 9: Histogram of sampled Gamma distribution against the analytical PDF ($n = 25$, $\lambda = 2$)

3.4 Beta

Yet another distribution that is also closely related to that of the Gamma, Exponential, and Poisson is the Beta distribution. The PDF of the Beta distribution, for $0 \leq x \leq 1$ and $\alpha, \beta > 0$ is defined as

$$f(x; \alpha, \beta) = \frac{1}{B(\alpha, \beta)} x^{\alpha-1}(1-x)^{\beta-1}, \tag{33}$$

where $B(\alpha, \beta) = \frac{\Gamma(\alpha)(\beta)}{\Gamma(\alpha+\beta)}$ [12].

Just as with the Gamma distribution, directly applying the ITM is not recommended since integrating the PDF of Eq. (33) is rather challenging. However, it is possible to find a relation between the Beta and Gamma distribution, which allows to generate a sample of the Beta distribution without using integration.

Consider the two random independently distributed variables, $X \sim Gamma(a, \beta)$ and $Y \sim Gamma(b, \beta)$ Now consider the random variable V as the sum of these two random variables. Thus, $V = X + Y$. Also, consider W as the fraction of X over the sum of X and Y, thus $W = \frac{X}{X+Y}$. Now, rewrite these equations for X and Y in terms of V and W:

$$\begin{aligned} \frac{X}{X+Y} &= W \\ \frac{X}{V} &= W \\ X &= VW \end{aligned} \tag{34}$$

and

$$\begin{aligned} X + Y &= V \\ VW + Y &= V \\ Y &= V(1-W). \end{aligned} \tag{35}$$

Using this result, the joint PDF of V and W can be defined using a 2D transformation on the joint PDF of X and Y:

$$f_{V,W}(v, w) = f_{X,Y}(x(v, w), y(v, w))||J||, \tag{36}$$

where $||J||$ is the absolute value of the determinant of the Jacobian matrix

$$\begin{aligned} |J| = \left|\left| \begin{matrix} \frac{\partial(x,y)}{\partial(v,w)} \end{matrix} \right|\right| &= \left|\left| \begin{matrix} \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \end{matrix} \right|\right| \\ &= \left|\left| \begin{matrix} \frac{\partial(vw)}{\partial v} & \frac{\partial(vw)}{\partial w} \\ \frac{\partial(v(1-w))}{\partial v} & \frac{\partial(v(1-w))}{\partial w} \end{matrix} \right|\right| \\ &= \left|\left| \begin{matrix} w & v \\ 1-w & -v \end{matrix} \right|\right| \\ = |-vw - v(1-w)| = |-vw + vw - v| &= |-v| = v. \end{aligned} \tag{37}$$

Furthermore, from the fact X and Y are statistically independent, the joint PDF of these two random variables can be rewritten as

$$f_{X,Y}(x(v,w), y(v,w)) = f_X(x(v,w)) \cdot f_Y(y(v,w)), \tag{38}$$

and using the definition of the PDF of the Gamma distribution, this equation becomes

$$\begin{aligned} f_{X,Y}(x,y) &= \frac{\beta^a}{\Gamma(a)} x(v,w)^{a-1} e^{-\beta x(v,w)} \cdot \frac{\beta^b}{\Gamma(b)} y(v,w)^{b-1} e^{-\beta y(v,w)} \\ &= \frac{\beta^a}{\Gamma(a)} (vw)^{a-1} e^{-\beta vw} \cdot \frac{\beta^b}{\Gamma(b)} (v(1-w))^{b-1} e^{-\beta v(1-w)} \end{aligned} . \tag{39}$$

Inserting Eq. (37) and (39) into Eq. (36) yields:

$$\begin{aligned} &\frac{\beta^a}{\Gamma(a)} (vw)^{a-1} e^{-\beta vw} \cdot \frac{\beta^b}{\Gamma(b)} (v(1-w))^{b-1} e^{-\beta v(1-w)} \cdot v \\ &\frac{\beta^{a+b}}{\Gamma(a)\Gamma(b)} v^{a-1} w^{a-1} e^{-\beta vw - \beta v(1-w)} v^{b-1} (1-w)^{b-1} v \\ &\frac{\beta^{a+b}}{\Gamma(a)\Gamma(b)} v^{a+b-1} w^{a-1} e^{-\beta v} (1-w)^{b-1} \end{aligned} . \tag{40}$$

The next step would be to find the marginal PDF of W. This can be found by integrating the joint PDF over the whole range of V, which is from zero to infinity as this is the range for which the gamma distribution is defined [10]. Thus,

$$\begin{aligned} f_W(w) &= \int_{v=0}^{\infty} \frac{\beta^{a+b}}{\Gamma(a)\Gamma(b)} w^{a-1} v^{a+b-1} e^{-\beta v} (1-w)^{b-1} dv \\ &= \frac{\beta^{a+b}}{\Gamma(a)\Gamma(b)} w^{a-1} (1-w)^{b-1} \int_{v=0}^{\infty} v^{a+b-1} e^{-\beta v} dv \end{aligned} . \tag{41}$$

The integral can be solved by substituting $u = \beta v$. This implies that, $du = \beta dv$ and thus $v = u/\beta$. Using this substitution, the integral becomes

$$\frac{1}{\beta^{a+b}} \int_0^{\infty} u^{a+b-1} e^{-u} du \tag{42}$$

and this can be recognized as the Gamma function with the argument $a+b$, as shown in Eq. (29). Therefore, Eq. (41) can be rewritten as

$$\begin{aligned} &\frac{\beta^{a+b}}{\Gamma(a)\Gamma(b)} w^{a-1} (1-w)^{b-1} \beta^{-a-b} \Gamma(a+b) \\ &= \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} w^{a-1} (1-w)^{b-1} \\ &= \frac{1}{B(a,b)} w^{a-1} (1-w)^{b-1} \end{aligned} . \tag{43}$$

As can be seen, this result matches with the definition of the PDF of the Beta distribution shown in Eq. (33) [13] .
It can thus be concluded that W follows as beta distribution with parameters a and b . Using this result, it is possible to generate a sample of the Beta distribution. Firstly, the samples of X and Y that follow a Gamma distribution, with parameters α, θ and β, θ respectively, are generated by the techniques discussed in section 3.3. The sample W of the Beta distribution with parameters α, β can then be computed by simply taking the fraction of X over $X+Y$. Figure 10 below shows a random sample of the Beta distribution generated by using the above method.

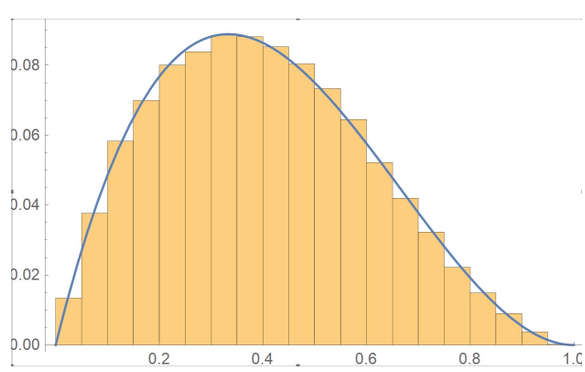

Figure 10: Histogram of the sampled Beta distribution against the analytical PDF ($\alpha = 2, \beta = 3$)

3.5 Chi Square

The Chi-Square distribution is mainly used in hypothesis testing and the creation of confidence intervals for the variance in Normal distributions. It is defined as the sum of ν squared independently distributed standard Normal variables, and therefore ν is also called the degrees of freedom or order of the Chi-Square distribution [14]. The PDF of this distribution is given as

$$f(x;\nu) = \frac{x^{\frac{\nu}{2}-1}e^{-\frac{x}{2}}}{2^{\frac{\nu}{2}}\Gamma(\frac{\nu}{2})}, \quad 0 < x < \infty \tag{44}$$

Comparing Eq. (44) and (28), it is rather trivial to see that the Chi-Square distribution is just a special case of the Gamma distribution, namely one where the shape and scale parameter are set to be $\frac{\nu}{2}$ and 2 respectively. Recall the results obtained in section 3.2 and 3.3: the random variable $X \sim Gamma(\alpha = n, \beta = \lambda)$ can be sampled by summing the random variables $Y_1 \dots Y_n$, which are independently distributed and follow an Exponential distribution with rate parameter λ. Furthermore, $Y_1 \dots Y_n$ can be sampled using the inverse CDF of the Exponential distribution as described in section 3.2. All of this implies that a sample of the random variable $Z \sim \chi^2(k)$ can be generated by taking the sum of $Y_1 \dots Y_{\frac{\nu}{2}}$, which all follow an Exponential distribution with rate parameter $\lambda = \beta = \frac{1}{\theta} = \frac{1}{2}$. Figure 11 below shows an example of a random sample of the Chi-Squared distribution, using the above result.

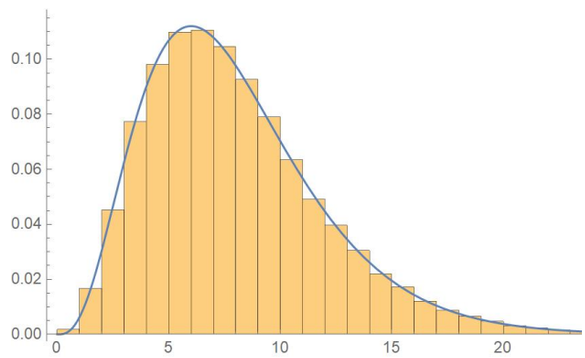

Figure 11: Histogram of the sampled Chi-Square distribution against the analytical PDF ($\nu = 8$)

4 Box-Mueller Transformation

The Box-Mueller Transform (BMT) is a method designed specifically to generate a sample of the standardized Normal distribution. The ITM method described in section 3 will be computationally insufficient to generate a sample of this kind of distribution since the CDF of the Normal distribution is rather hard to define as it will involve the use of the error function [15].

The derivation of the BMT starts by defining two independent variables that follow a standard normal distribution: $X = Y \sim N(0,1)$. The joint probability density function will thus be given as the product of the PDFs of X and Y:

$$\begin{aligned} f_{X,Y}(x,y) = f_X(x) \cdot f_Y(y) &= \frac{1}{\sigma_x\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x-\mu_X}{\sigma_X}\right)^2\right) \frac{1}{\sigma_Y\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{y-\mu_Y}{\sigma_Y}\right)^2\right) \\ &= \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}x^2\right) \cdot \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}y^2\right) \\ &= \frac{1}{2\pi} \exp\left(-\frac{1}{2}\left(x^2+y^2\right)\right) \quad . \end{aligned} \tag{45}$$

A visual representation of this joint PDF is created using *Mathematica* and is shown in figure 12 below. Consider an arbitrary X, Y plane of this joint PDF. Figure 13 gives a visual representation of this plane. As can be seen, this plane contains a number of circles that are centered around the origin.

Imagine a line segment that connects the origin of the joint PDF with any point (x, y) on one of the circles. Using the Pythagorean theorem, the length, r, of this line segment can be denoted as

$$r = \sqrt{x^2 + y^2} \tag{46}$$

or

$$r^2 = x^2 + y^2. \tag{47}$$

I.e., The random variable R^2 can be seen as the sum of two squared standardized normal distributions. Now, recall the definition of the Chi-Squared distribution. If $M_1, M_2, \dots, M_K$ are independent random variables following a standardized normal distribution, then the sum of their squares

$$Q = \sum_{i=1}^{k} M_i^2 \tag{48}$$

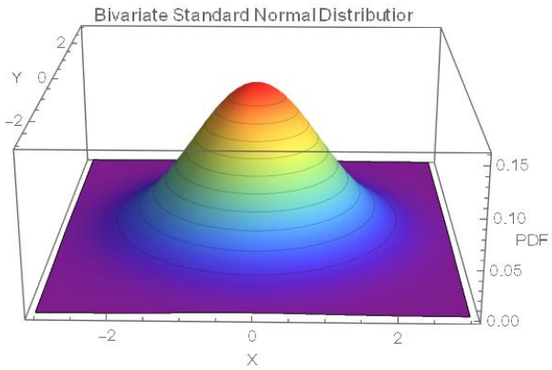

Figure 12: 3D Plot of the bivariate standard Normal distribution.

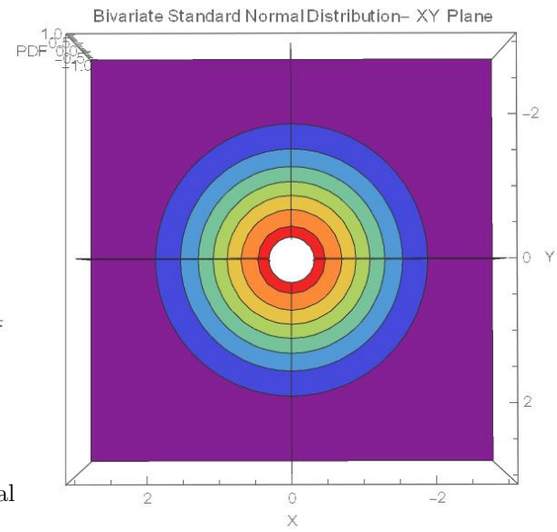

Figure 13: X, Y-plane of the bivariate standardized Normal distribution at $Z = 1$

will have a Chi-Squared distribution with k degrees of freedom. Thus $Q \sim \chi^2(k)$ [14]. In section 3.5, it is proven that the Chi-Squared distribution is a special case of the Gamma distribution where the shape and scale parameter are $\frac{k}{2}$ and 2 respectively. As a consequence, this means that the Chi-Squared distribution is the same as the sum of $\frac{k}{2}$ Exponential distributions with the rate parameter $\lambda = \frac{1}{2}$. Furthermore, the Exponential distribution can be rewritten in terms of its inverse CDF as shown in Eq. (26). Therefore, the PDF of Q can be formulated as

$$\begin{aligned} f_Q(q) &= \sum_{i=1}^{\frac{k}{2}} q_i = -\sum_{1}^{\frac{k}{2}} \frac{\ln(1-u)}{\frac{1}{2}} \\ &= -\frac{k}{2}\left(2\ln(1-u)\right) = -k\ln(1-u) \\ &= -k\ln(u), \end{aligned} \tag{49}$$

where $u \sim U_1(0,1)$. In the case of the random variable R^2, k is equal to 2, as this random variable describes the sum of squares of two standardized normal distributions. Rewriting the equation for R yields

$$R = \sqrt{-2\ln(U_1)}. \tag{50}$$

The angle, θ, between the line segment and the positive X direction is defined as

$$\theta = \arctan\left(\frac{Y}{X}\right). \tag{51}$$

Note, that all the points on these circles have an equal height, i.e. they denote an equal probability. Therefore, $\theta \sim U_2(0, 2\pi)$, which can be transformed rewritten as $\theta \sim 2\pi U_2(0,1)$. Note, that U_1 and U_2 are statistically independent from each other.

Now both the absolute value and the argument of the points on the joint PDF of the two standardized normal distributions are defined. The current situation of the above analysis is summarized in figure 14 below.
Using these results in combination with basic geometry, it follows that the X and Y coordinates of the point on the circle are given as

$$X = R\cos(\theta) = \sqrt{-2\ln(U_1)}\cos(2\pi U_2) \tag{52}$$

and

$$Y = R\sin(\theta) = \sqrt{-2\ln(U_1)}\sin(2\pi U_2). \tag{53}$$

X and Y are both random variables following a standardized Normal distribution [16, 17]. Both equations can be used to generate a sample of the standardized Normal distribution. Figure 15 below shows the resulting sampled distribution using Eq. (52).

The next step would be to transform this sample of the standardized normal distribution to a non-standardized form using Z-scores. Recall the definition of the Z-scores as

$$Z = \left(\frac{X-\mu}{\sigma}\right). \tag{54}$$

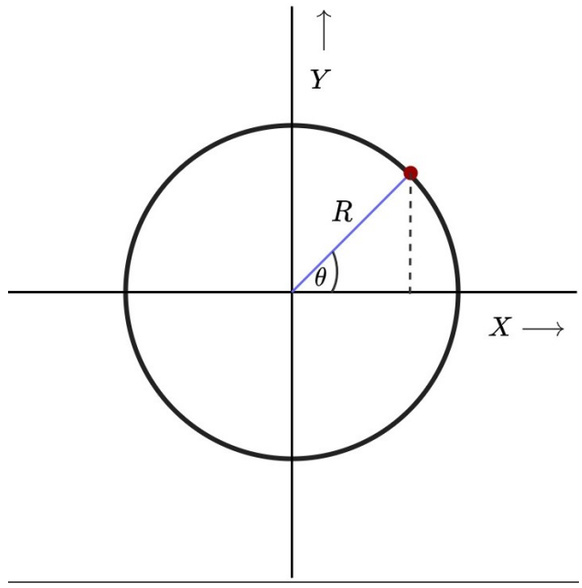

Figure 14: Overview of the known quantities in the XY plane of the bivariate Normal distribution

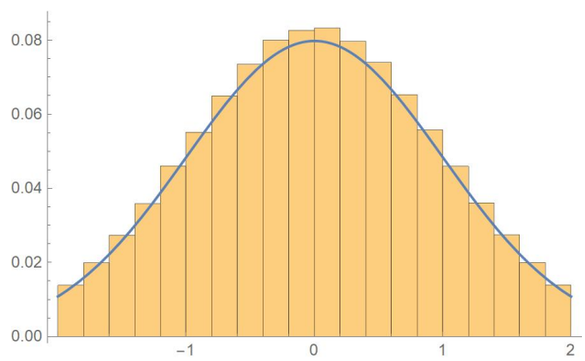

Figure 15: Histogram of the sampled standardized Normal distribution against the analytical PDF.

Here, Z is the standard Normal variable and X is the non-standard Normal variable with mean μ and standard deviation σ. Therefore,

$$X = \sigma Z + \mu, \tag{55}$$

which implies that going from the standard Normal distribution found using the BMT to the non-standard form, only requires multiplication by the desired standard deviation and then adding the mean. Figure 16 shows an example of a sampled distribution using this transformation.

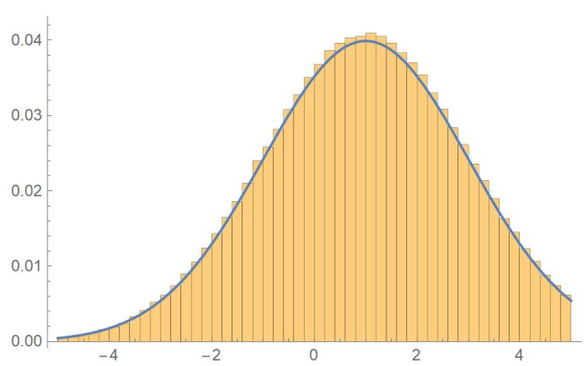

Figure 16: Histogram of a sampled Normal distribution in non-standard form against the analytical PDF ($\mu = 1$, σ =2).

5 Accept and Reject Method

In addition to the sampling methods described in the previous sections, the Acceptance and Rejection Method (ARM) can also be used the generate a sample from a specified distribution. In general, the idea of this method is to generate observations from a proposed distribution and then accept or reject each of these observations based on whether or not they could be part of the desired sample.

Suppose that $f(x)$ denotes the target PDF that needs to be sampled, but it is rather challenging to do so. The ARM takes care of this problem by implementing the following procedure [18]:

1. Choose another arbitrary PDF, $g(x)$ called the proposal distribution, which is easier to sample from than the target distribution. Furthermore, a constant c is defined and $g(x)$ is chosen, such that

$$f(x) \leq c \cdot g(x) \tag{56}$$

This implies that $c \cdot g(x)$ has to completely envelop the target distribution.

2. Rewriting Eq. (56) to free c yields

$$\begin{aligned} \frac{f(x)}{g(x)} &\leq c \\ c &\geq \frac{f(x)}{g(x)} \end{aligned}, \tag{57}$$

which implies that c has to be at least equal to the maximum value of the fraction $\frac{f(x)}{g(x)}$.

3. Denote the random variable $Y \sim g(x)$ and take n observations of this distribution. A single observation of this distribution is called y_i . Now the acceptation rule is defined as

$$\frac{f(y_i)}{c \cdot g(y_i)} \leq 1, \tag{58}$$

meaning that the proposed sample y_i is accepted if it meets the above condition. Similarly, n observations from the unity Uniform distribution can be drawn and each observation is called u_i. This implies that the sample y_i is accepted when $u_i \leq \frac{f(y_i)}{c \cdot g(y_i)}$ and rejected otherwise. This last result is also called the Acceptance-Rejection rule

The working of this algorithm can be verified by showing that the distribution of the accepted samples is equal to that of the desired distribution. The first step of this proof is to compute the rejection rate using the laws of conditional probability [18]:

$$\begin{aligned} P(Accepted) &= \sum_i P[Accept|Y = y_i]P[DrawY = y_i] \\ &= \sum_i P[c \cdot u \cdot g(y_i \leq f(y_i)|Y = y_i g(y_i)] \\ &= \sum_{y_i} \frac{f(y_i)}{c \cdot g(y_i)} g(y_i) \\ &= \sum_i \frac{f(y_i)}{c} = \frac{1}{c}. \end{aligned} \tag{59}$$

When inserting this result into Bayes' Theorem, it is possible to calculate the probability of the value of the y_i given that it is accepted:

$$\begin{aligned} P(Y = y_i|Accept) &= \frac{P(Y = y_i \cap Accept)}{P(Accept)} = \frac{P(Accept|Y = y_i)P(Draw\ y_i)}{P(Accept)} \\ &= \frac{P(Accept|Y = y_i)g(y_i)}{\left(\frac{1}{c}\right)} = \frac{f(y_i)}{c \cdot g(y_i) c \cdot g(y_i)} \\ &= f(y_i) \end{aligned} \quad . \tag{60}$$

The ARM can thus be used for generating samples of almost any distribution, but the conditions are that the target PDF does not go to infinity, and that it should be relatively easy to determine the peak value. The latter condition is to ensure that proposed PDF covers the whole target.

The downside, however, is the inefficiency of the algorithm. It can be rather difficult to find such a combination for c and $g(x)$ that has relatively little area, but still covers the whole of the target distribution. Furthermore, the algorithm only uses a portion of the samples generated as not all of them are accepted.

In the coming sections, a number of examples are given to show the performance of the ARM method.

5.1 Exponential

In contrast to the previous chapters, the example discussed below will not be solved symbolically, as this method requires some visual information that changes for each value of the λ-parameter. Instead, a worked-out example will be given which describes the necessary steps that can be used for every other form of the Exponential distribution.

Consider that the target distribution that is given by $X \sim Exponential(4)$. Figure 17 shows the density function of this corresponding PDF.
As can be seen, it is rather unlikely that $X \leq 2$. Therefore, the analysis will only focus on the range $0 \leq X \leq 2$ and ignore the other values. A normalized uniform distribution with the same range can be used for the proposed distribution. Thus,

$$g(x) = \frac{1}{2} \quad 0 \leq x \leq 2, \tag{61}$$

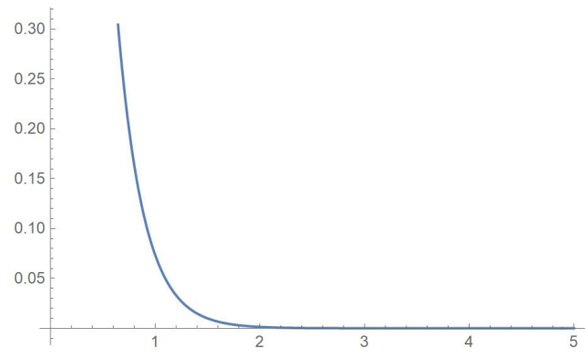

Figure 17: Plot of the Exponential PDF with rate parameter $\lambda = 4$

and this implies that the constant c is then determined as

$$\begin{aligned} c &\geq \frac{f(x)}{g(x)} \\ c &\geq \frac{4e^{-4x}}{\frac{1}{3}} \\ c &\geq 12e^{-4x}, \end{aligned} \tag{62}$$

which simplifies to

$$c = Max\left[12e^{-4x}\right]. \tag{63}$$

Usually, such an expression can be determined by setting the first derivative equal to zero and solving for x. In this case, that solution would yield $x \longrightarrow -\infty$, but since the lower bound of the Exponential distribution is at $x = 0$, this value is used instead. Therefore,

$$c = 12e^{-4\cdot 0} = 12. \tag{64}$$

Using these known values for c and $g(x)$, the Accept and Reject Sampling technique can be performed with the accept condition being

$$u_i \leq \frac{f(x_i)}{g(x_i)}, \tag{65}$$

where u_i and x_i are generated observations of the unity uniform and proposal distribution respectively. Figure 18 below shows the result of a generated sample of this Exponential distribution consisting out of 50 000 observations.

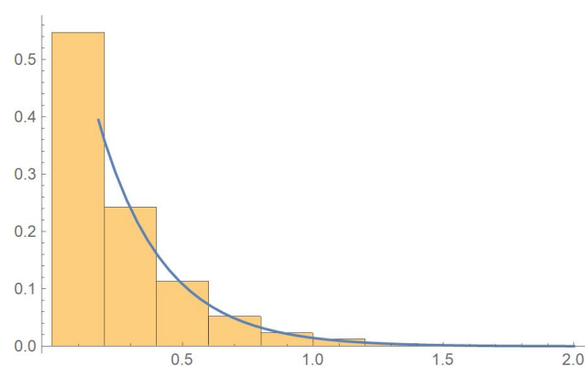

Figure 18: Histogram of the sampled Exponential distribution against the analytical PDF ($\lambda = 4$)

5.2 Gamma

Now consider that the random variable X follows the target distribution which is a Gamma distribution as defined in section 3.3. Figure 19 shows the PDF of an example of this distribution.
As shown, the probability is relatively small at $X \approx 3$. However, even if X increases further, the probability never becomes zero. Let's set the upper bound (up) of this analysis equal to the value of x for which by setting $f_X(x) = 0.001$. Note that this equation will yield two solutions: one near $x \approx 0$ and the other depends on the value of the distribution parameters. For the value of up, the latter solution is used. For the rest of this problem, only the domain of $0 \leq X \leq up$ is considered and the other values are ignored. The proposed distribution will use a normalized uniform distribution with the same range. Thus,

$$g(x) = \frac{1}{up}. \tag{66}$$

The value for the constant c is again equal to the maximum value of the fraction of $f_X(x)$ over $g(x)$, which can be found using *Mathemtica's* built-in commands.
Now these values are found, the ARM can be used. Figure 20 below shows the result of a generated sample of this Gamma distribution consisting out of 50 000 observations.

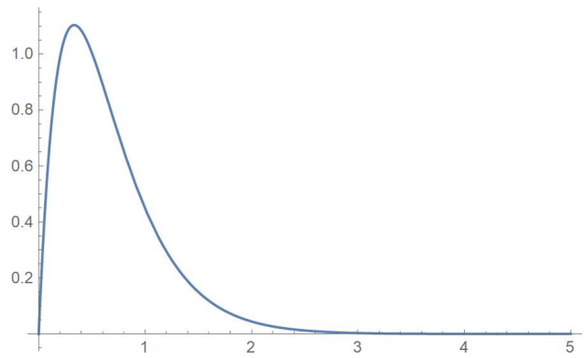

Figure 19: Plot of the Gamma PDF ($\alpha = 2$, $\beta = \lambda = 3$)

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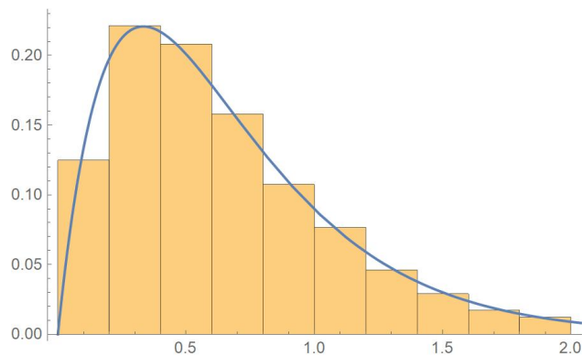

Figure 20: Histogram of the sampled Gamma distribution against the analytical PDF ($\alpha = 2$, $\beta = \lambda = 3$)

5.3 Beta

The Beta distribution is just a special case of the Gamma distribution and sampling it using the ARM is rather straightforward. Recall that the Beta distribution is only defined for $0 \leq x \leq 1$. This means that the proposed distribution can be given by a normalized uniform distribution with a range of (0,1) (which is just the same as the unity Uniform distribution). I.e.,

$$g(x) = 1. \tag{67}$$

This means that the constant c is just equal to the maximum value of the Beta distribution, which can be found by using the in-built commands of *Mathematica*. Now, all the quantities are defined to use the ARM. Figure 21 below shows the result of a generated sample of the Beta distribution consisting out of 50 000 observations

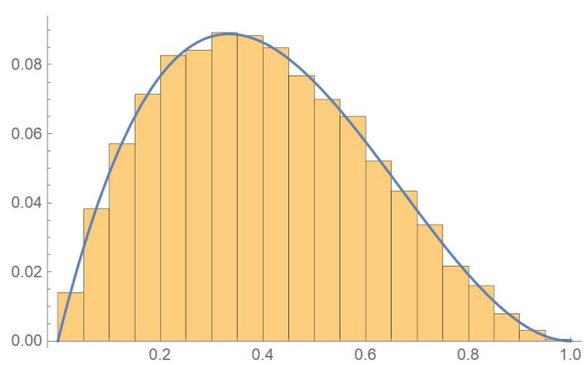

Figure 21: Histogram of the sampled Beta distribution against the analytical PDF ($\alpha = 2$, $\beta = \lambda = 3$)

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5.4 Polynomial

The Polynomial distribution has a PDF in the form of, as the name already indicates, a polynomial of an arbitrary degree (n). Consider a random variable $X \sim Poly(a_n, a_{n-1}, \ldots a_1, a_0)$, then the PDF of this random variable is given as

$$f_X(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0, \tag{68}$$

where $a_n, a_{n-1} \ldots a_1, a_0$ are the coefficients of the power terms in the polynomial. Note that since the Polynomial distribution is strictly increasing, the area under the curve will be infinite as $x \longrightarrow \infty$. Therefore, an upper bound (up) for the function needs to be defined such that it can be normalized by dividing over the area underneath it (A), which is given by

$$A = \int_0^{up} a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 dx. \tag{69}$$

The normalized PDF of the random variable X is thus given as

$$f_X(x) = \frac{a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0}{A}. \tag{70}$$

The proposed distribution follows thus again as

$$g_X(x) = \frac{1}{up} \tag{71}$$

, and the value for c can be found by

$$\begin{aligned} c &= Max\left[\frac{f_X(x)}{g_X(x)}\right] \\ &Max\left[\frac{\left(\frac{a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0}{A}\right)}{\left(\frac{1}{up}\right)}\right] \\ &= \frac{up}{A} Max\left[a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0\right] \end{aligned} \tag{72}$$

Note that the last step is justified since the values for up and A are constants. Furthermore, note that $Max\left[a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0\right]$ is simply the value for $f(up)$ as the PDF is strictly increasing. All quantities are known in order to perform the ARM. Figure 22 below shows an example of the generation of a random Polynomial distribution using the ARM.

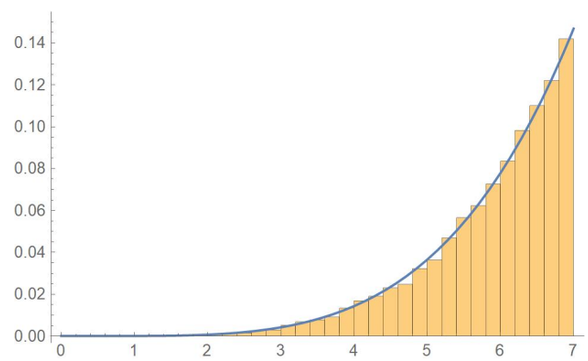

Figure 22: Histogram of the sampled Polynomial distribution against the analytical PDF ($a_4 = 4$, $a_3 = -2$, $a_2 = -3$, $a_1 = 1$, $a_0 = 5$)

5.5 Weibull

The Weibull distribution is a continuous probability distribution that is mainly used for modeling the time to failure or the time between two events. The PDF of a random variable X with this distribution is given as

$$f_X(x; k, \lambda, \theta) = \frac{k}{\lambda}\left(\frac{x-\theta}{\lambda}\right)^{k-1} e^{-\left(\frac{x-\theta}{\lambda}\right)^k} \quad x \geq \theta, \tag{73}$$

where $k, \lambda > 0$ are the scale and shape parameters respectively, and θ is the location parameter [19]. Figure 23 below shows the PDF of an example of this distribution. Just as in some of the previous examples, the

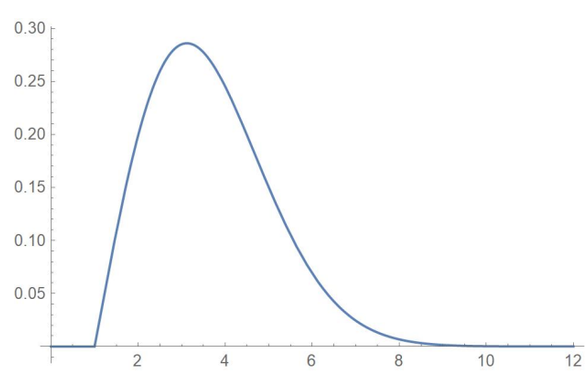

Figure 23: Plot of the Weibull PDF ($k = 2$, $\lambda = 3$, $\theta = 1$)

PDF is approaching zero for the value of the random variable is getting larger and larger. Therefore, the value for up is again equal to the largest value of x for which $f_X(x) = 0.001$. The domain is thus only defined for $\theta \leq X \leq up$. Therefore, the proposed distribution is

$$g_X(x) = \frac{1}{up - \theta}. \tag{74}$$

Furthermore, the value for c is given as

$$c = Max\left[\frac{f_X(x)}{\left(\frac{1}{up-\theta}\right)}\right] = (up-\theta)\,Max\left[f_X(x)\right], \tag{75}$$

where the value of $Max[f_X(x)]$ can be found using *Mathematica's* built-in commands.

All quantities are now known to determine the accept condition and, the ARM can thus be performed. Figure 24 below show the generated random sample of a Weibull distribution consisting out of 50 000 observations.

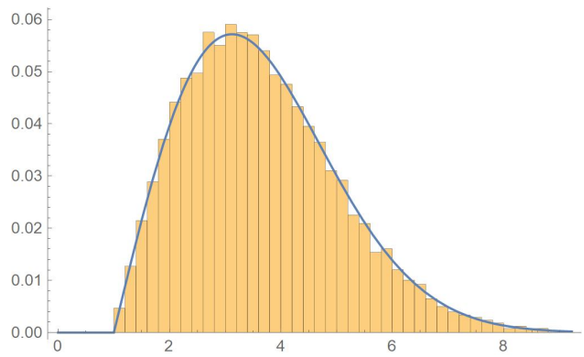

Figure 24: Histogram of the sampled Weibull distribution against the analytical PDF ($k = 2$, $\lambda = 3$, $\theta = 1$)

6 Conclusion and Discussion

In conclusion, there are various ways to generate random samples of almost any desired distribution, given that it is rather easy for computers to generate a random sample of the Uniform distribution.

Relatively simple distributions can be sampled using Additive sampling, which basically means that a simple transformation of the base distribution, e.g. the Uniform distribution, is repeatedly added or subtracted to get closer and closer to the desired distribution. This method is rather intuitive and easy to explain, but it cannot always be implemented as it will require a formulation for the necessary transformation, which can sometimes be hard to find.

The Inverse Transform Method is an algorithm that transforms data from the Uniform distribution into that of any other desired distribution by making use of the inverse CDF. Although it is also a relatively efficient method, it cannot be used for PDFs that are not strictly increasing. Furthermore, this method can be hard to use for distributions that have PDFs that are difficult to integrate. However, when combined with Additive sampling this latter limitation can be resolved in some cases.

The Box-Mueller Transform is a method designed specifically for generating random samples of the standardized Normal distribution. It makes use of the symmetry of the joint PDF of the bivariate Normal distribution, along with some properties of the Chi-Square distribution.

Lastly, the Accept and Reject Method is introduced. This method works by first defining a proposal distribution, which is usually a scaled and/or offset form of the Uniform distribution, and then accepts a fraction of these observations based upon the defined acceptance rule. Doing so will transform the proposed distribution into the target distribution. The advantage of this method, compared to that of the ITM and Additive sampling, is that it can be used for creating a sample of all most every distribution. However, this method is computationally intensive and relatively inefficient as a substantial amount of the generated samples are rejected. Something which is not the case for the other methods that are investigated, which generally utilize all the generated samples as it simply transforms them.

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